

Lab 2 Deliverable: Agile Backlog Epics & User Stories

Problem Statement #40 — Student Startup Crowdfunding Platform | Submitted by: Sambhram M (SRN: PES1UG24CS409)

1. Functional Requirement to Agile Epic Decomposition

In accordance with Agile development practices, the 5 core Functional Requirements identified in Lab 1 (FR-001 through FR-005) have been systematically decomposed into 5 thematic Epics and 12 granular User Stories. Each User Story articulates the specific persona, feature goal, and business value using standard Agile framing, accompanied by strict Acceptance Criteria and Fibonacci effort estimation.

Epic ID & Theme	Story ID	User Story (As a... I want... So that...)	Acceptance Criteria	Priority	Pts	Sprint
Epic 1: Campaign Creation (FR-002, UC-01)	SSCP-1	As a Student Founder, I want to define my campaign title, pitch, category, and total funding target, so that potential backers understand the startup proposition and capital needs.	Pass: Valid venture title, target (>0), description initialize draft. Fail: Null title or non-positive target.	High	5	Sprint 1
Epic 1: Campaign Creation (FR-002, UC-01)	SSCP-2	As a Student Founder, I want to set the start and end dates for my campaign, so that the fundraising campaign is time-bound and builds momentum.	Pass: Campaign duration between 14-60 days validated. Fail: End date before start date or duration > 60 days.	High	3	Sprint 1
Epic 1: Campaign Creation (FR-002, UC-01)	SSCP-3	As a Student Founder, I want to divide my project roadmap into distinct sequential milestones with individual funding tranches, so that funds can be disbursed in structured, verifiable phases.	Pass: Sum of milestone tranches equals target; each milestone has deliverables & due date. Fail: Tranche sum mismatch.	High	8	Sprint 1
Epic 2: Backer Pledging & Escrow (FR-003, UC-02, UC-03)	SSCP-4	As a Campaign Backer, I want to view available reward tiers and select one (or opt out) during checkout, so that I receive tangible incentives and perks for my contribution.	Pass: Chosen reward tier captured and printed on confirmation receipt. Fail: Pledge accepted without recording reward tier state.	Medium	5	Sprint 1
Epic 2: Backer Pledging & Escrow (FR-003, UC-02, UC-04)	SSCP-5	As a Campaign Backer, I want to authorize my payment via a PCI-DSS compliant gateway into platform escrow, so that my funds are protected until milestones are formally verified.	Pass: Tokenized payment authorized and funds locked in campaign escrow. Fail: Direct disbursement without escrow hold.	High	8	Sprint 1
Epic 3: Milestone Review System (FR-004, UC-05, UC-06)	SSCP-6	As a Student Founder, I want to upload completion evidence (code repos, demo links, reports) for a due milestone, so that the faculty committee can evaluate actual progress.	Pass: Deliverable links/files attached; milestone advances to 'Under Review'. Fail: Evidence submitted for past/inactive milestones.	High	5	Sprint 2
Epic 3: Milestone Review System (FR-004, UC-06)	SSCP-7	As a Faculty Committee Member, I want to inspect submitted deliverables and record an Approve or Reject decision with comments, so that fund disbursement is justified by verified work.	Pass: Decision, reviewer ID, timestamp, and audit comments logged immutably. Fail: Milestone status altered without reviewer credentials.	High	5	Sprint 2
Epic 3: Milestone Review System (FR-004, UC-06)	SSCP-8	As a Student Founder, I want to review faculty rejection feedback and resubmit revised evidence before the deadline, so that I can rectify deficiencies without forfeiting the campaign.	Pass: Resubmission within grace window resets state to 'Under Review'. Fail: Resubmission attempted past grace window.	Medium	3	Sprint 2

Epic ID & Theme	Story ID	User Story (As a... I want... So that...)	Acceptance Criteria	Priority	Pts	Sprint
Epic 4: Tranche Disbursement (FR-001, UC-07, UC-04)	SSCP-9	As a System Service, I want to verify that the escrow balance is sufficient and conditions are satisfied prior to release, so that fund releases never exceed collected pledges.	Pass: System verifies escrow balance $\geq$ tranche before triggering payout API. Fail: Tranche payout triggered with insufficient escrow.	High	3	Sprint 2
Epic 4: Tranche Disbursement (FR-001, UC-07, UC-04)	SSCP-10	As a Student Founder, I want approved milestone funds to be transferred directly to my linked bank account, so that I have the required capital to execute the next project phase.	Pass: Automated tranche payout completed, milestone status set to 'Released', ledger updated. Fail: Payout executed without faculty approval sign-off.	High	8	Sprint 2
Epic 5: Automated Notifications & Ledger (FR-005, UC-08)	SSCP-11	As a Student Founder and Backer, I want to receive automated email and in-app alerts whenever a milestone is approved, rejected, or a tranche is released, so that I stay informed within 5 minutes of each event.	Pass: Automated alert dispatches within 5 minutes of triggering event. Fail: Milestone event occurs without triggering dispatch log.	Medium	5	Sprint 2
Epic 5: Automated Notifications & Ledger (FR-005, UC-08)	SSCP-12	As a Campaign Backer, I want to view a transparent milestone timeline and fund disbursement ledger, so that I can audit venture progress and fund allocation in real time.	Pass: Public dashboard reflects milestone verification timestamps and tranche disbursement amounts. Fail: Discrepancy between ledger and disbursed funds.	Low	3	Sprint 2

2. Planning Poker Estimation & Fibonacci Scale Rationale

Story points were assigned using the modified Fibonacci sequence (1, 2, 3, 5, 8, 13) via Planning Poker simulation. The Fibonacci sequence reflects non-linear increases in complexity, uncertainty, and cognitive overhead for larger tasks. Items estimated at 3 story points (SSCP-2, SSCP-8, SSCP-9, SSCP-12) reflect standard CRUD patterns and localized validations. Items at 5 story points (SSCP-1, SSCP-4, SSCP-6, SSCP-7, SSCP-11) involve cross-entity data persistence, role-based access control, and automated notification services. Items at 8 story points (SSCP-3, SSCP-5, SSCP-10) represent architectural complexity, including transactional financial escrow management, multi-tranche reconciliation, and external payment gateway webhooks.